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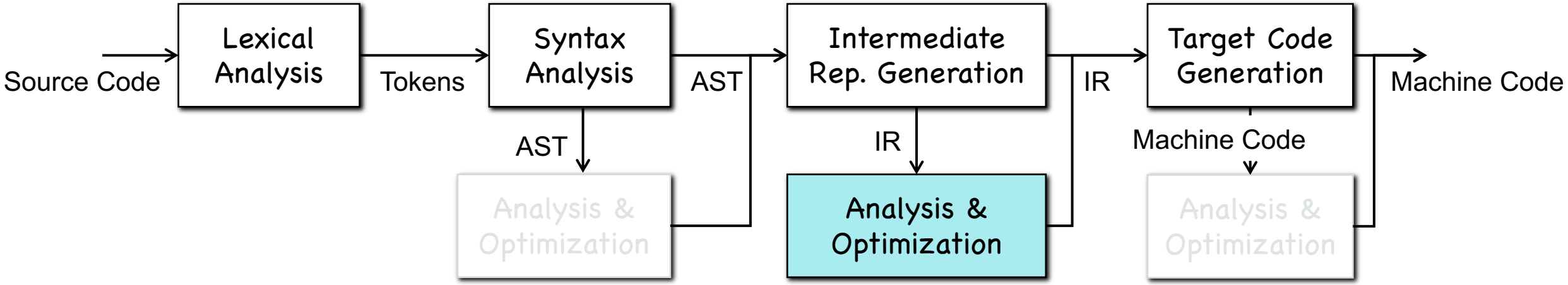
Chapter 12-2
Pointer Alias Analysis

Pointer Analysis



- Why pointer analysis?

```
*p = a; …; …; b = *q; // does b data-depend on a?
```

Pointer analysis is the most fundamental program analysis!

- Flow-sensitivity
- Context-sensitivity
- Path-sensitivity
- Field-sensitivity

Why Pointer Alias Analysis?

- **Aliases**: two expressions that denote the same memory location

- Aliases are introduced by
  - references/pointers
  - function call
  - array indexing
  - C unions
  - …

```
int x,y;
int *p = &x;
int *q = &y;
int *r = p;
int **s = &q;
```

Why Pointer Alias Analysis?

- Improving the precision of **compiler optimizations** that require knowing what is modified or referenced

```
x = 3;
*p = 4;
y = x; // can constant 3 propagate here
```

Constant Propagation

```
t = a + b;
*p = t
y = a + b; // is a + b available?
```

Available Expression

- Error detection

```
x.lock();
...
y.unlock(); // same object as x?
```

PART I: Pointer Alias Analysis

May/Must Pointer Alias Analysis

- **May- (Must-)** analysis checks if two expressions **may (must)** denote the same memory location.

- May analysis – Must-not analysis
  - If a *may-analysis* says 'no', it means two expressions must not denote the same memory location.
  - Useful in compiler optimization.

```
x = 3;
*p = 4;
y = x; // can constant 3 propagate here
```

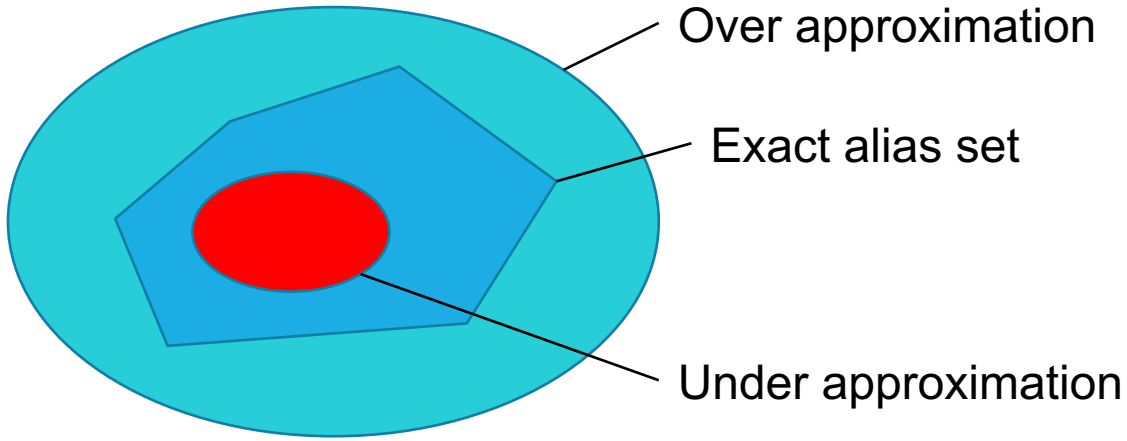
Constant Propagation

```
t = a + b;
*p = t
y = a + b; // a + b is available?
```

Available Expression

May/Must Pointer Alias Analysis

- Sound/Complete or Over-approximation/Under-approximation
- May analysis provides a sound (or over-approximated) alias set



The simplest sound alias analysis is to return true for all may-alias queries

- Tradeoff between soundness and precision (i.e., completeness)

Flow-Sensitive Pointer Analysis

- Flow-sensitive pointer analysis computes **for each program point** what memory locations a pointer expression may refer to
- Flow-sensitive pointer analysis is **(traditionally) too expensive** to perform for whole program
- There have been a few studies for flow-sensitive or even path-sensitive analysis

Flow-Sensitive Pointer Analysis

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- There have been a few studies for flow-sensitive or even path-sensitive analysis

[1] Falcon: A Fused Approach to Path-Sensitive Data Dependence Analysis
Peisen Yao, Jinguo Zhou, Xiao Xiao, Qingkai Shi, Rongxin Wu, and Charles Zhang, PLDI 2024.

[2] Value-Flow-Based Demand-Driven Pointer Analysis for C and C++
Yulei Sui and Jingling Xue, IEEE Transactions on Software Engineering 2018.

Flow-Insensitive Pointer Analysis

- Flow-insensitive pointer analyses for whole program analyses

Andersen's Algorithm
Presented by Andersen in 1994
Nearly cubic complexity

Steensgaard's Algorithm
Published in POPL 1996
Almost linear, O(nα(n))

A Small Pointer Language

- y = &x (address-taken)
- y = x (assignment)
- *y = x (store statement)
- y = *x (load statement)

- **pts(x)** --- the points-to set of x
  - pts(x) = { y, z } --- x may point to either y or z

Andersen's Algorithm

Inclusion-based pointer analysis

Statement	Constraint	Shorthand
y = &x	pts(y) ⊇ {x}	
y = x	pts(y) ⊇ pts(x)	
*y = x	∀ v ∈ pts(y): pts(v) ⊇ pts(x)	pts(*y) ⊇ pts(x)
y = *x	∀ v ∈ pts(x): pts(y) ⊇ pts(v)	pts(y) ⊇ pts(*x)

```
p  =  &a;
q  =  &b;
*p =  q;
r  =  &c;
s  =  p;
t  =  *p;
*s =  r;
```

?

```
p = &a;
q = p;
p = &b;
r = p;
```

```
pts(p) ⊇ { a };
pts(q) ⊇ pts(p);
pts(p) ⊇ { b };
pts(r) ⊇ pts(p);
```

```
pts(p) = { a, b };
pts(q) = { a, b };
pts(r) = { a, b };
pts(a) = pts(b) = {};
```

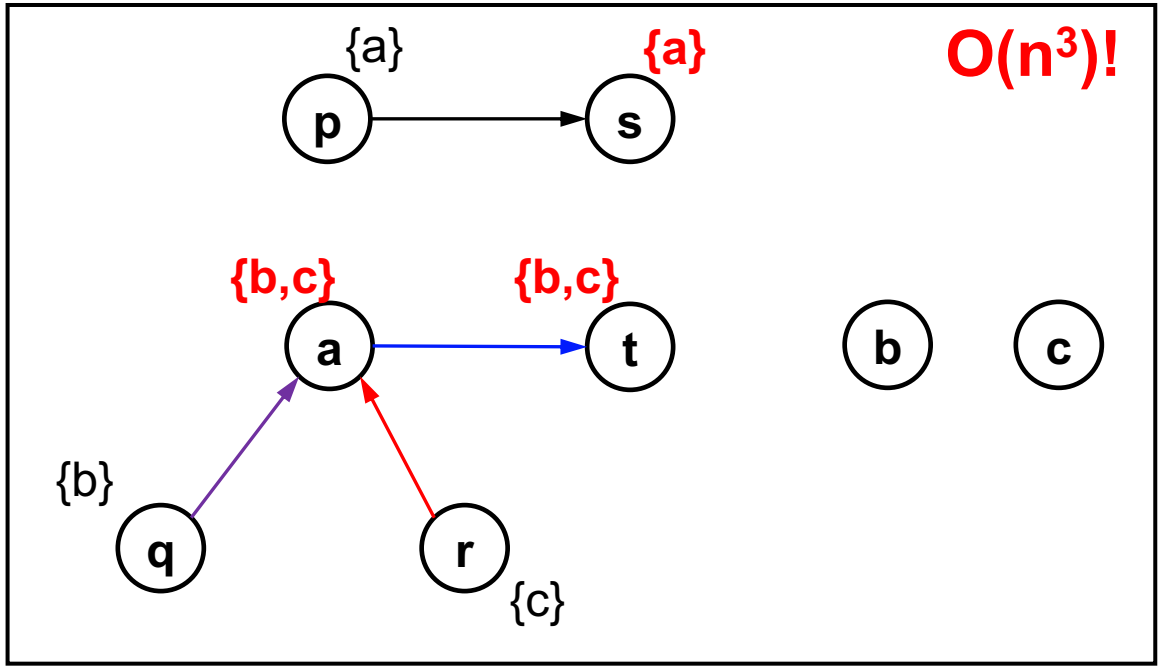
Over-approximation!
Sound result!

Andersen's Alg. as Graph Closure

- Each graph node x denotes the points-to set $pts(x)$
- Solving the set constraints via a dynamic transitive closure

```
p  =  &a;
q  =  &b;
*p =  q;
r  =  &c;
s  =  p;
t  =  *p;
*s =  r;
```

```
...
...
pts(*p) ⊇ pts(q)
...
...
pts(t) ⊇ pts(*p)
pts(*s) ⊇ pts(r)
```



Andersen's Alg. as Graph Closure

- Each graph node x denotes the points-to set $pts(x)$
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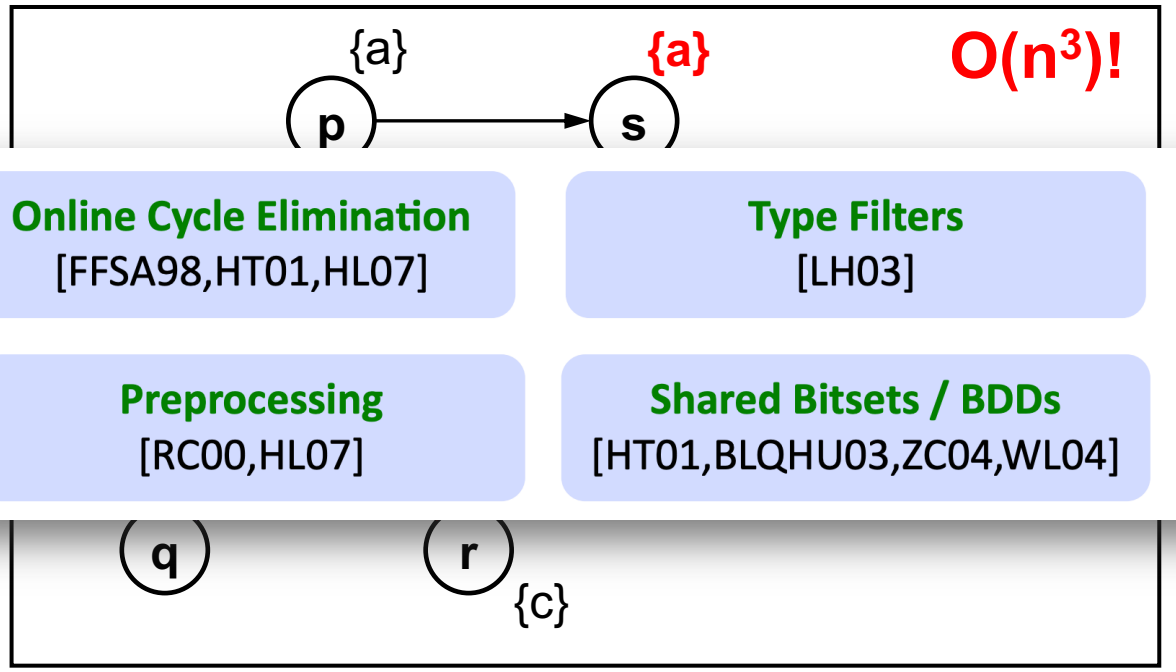
```
p  =  &a;
q  =  &b;
*p =  q;
r  =  &c;
s  =  p;
t  =  *p;
*s =  r;
```

```
...
...
pts(*p) ⊇ pts(q)
...
...
pts(t) ⊇ pts(*p)
pts(*s) ⊇ pts(r)
```



Online Cycle Elimination
[FFSA98,HT01,HL07]

Type Filters
[LH03]

Preprocessing
[RC00,HL07]

Shared Bitsets / BDDs
[HT01,BLQHU03,ZC04,WL04]

Steensgaard's Algorithm

Inclusion-based pointer analysis (Andersen's Algorithm)

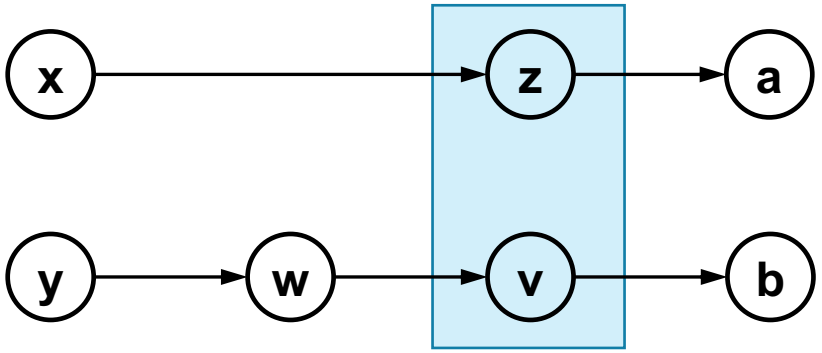
Statement	Constraint	Shorthand
y = &x	pts(y) ⊇ {x}	
y = x	pts(y) ⊇ pts(x)	
*y = x	∀ v ∈ pts(y): pts(v) ⊇ pts(x)	pts(*y) ⊇ pts(x)
y = *x	∀ v ∈ pts(x): pts(y) ⊇ pts(v)	pts(y) ⊇ pts(*x)

Unification-based pointer analysis (Steensgaard's Algorithm)

Statement	Constraint	Shorthand
y = &x	pts(y) ⊇ {x}	
y = x	pts(y) = pts(x)	
*y = x	∀ v ∈ pts(y): pts(v) = pts(x)	pts(*y) = pts(x)
y = *x	∀ v ∈ pts(x): pts(y) = pts(v)	pts(y) = pts(*x)

Steensgaard's Algorithm

- **Points-to graph**: x → y means y ∈ pts(x).

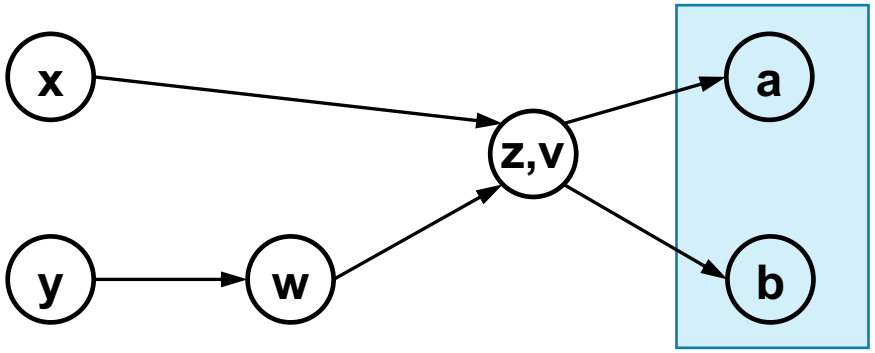


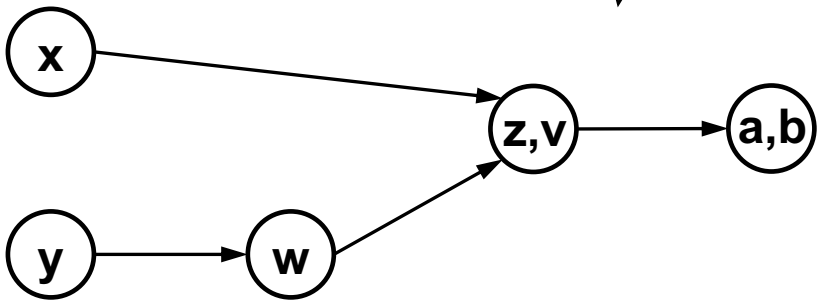
Steensgaard's algorithm: One node, one out-going edge!

⇦ x = *y

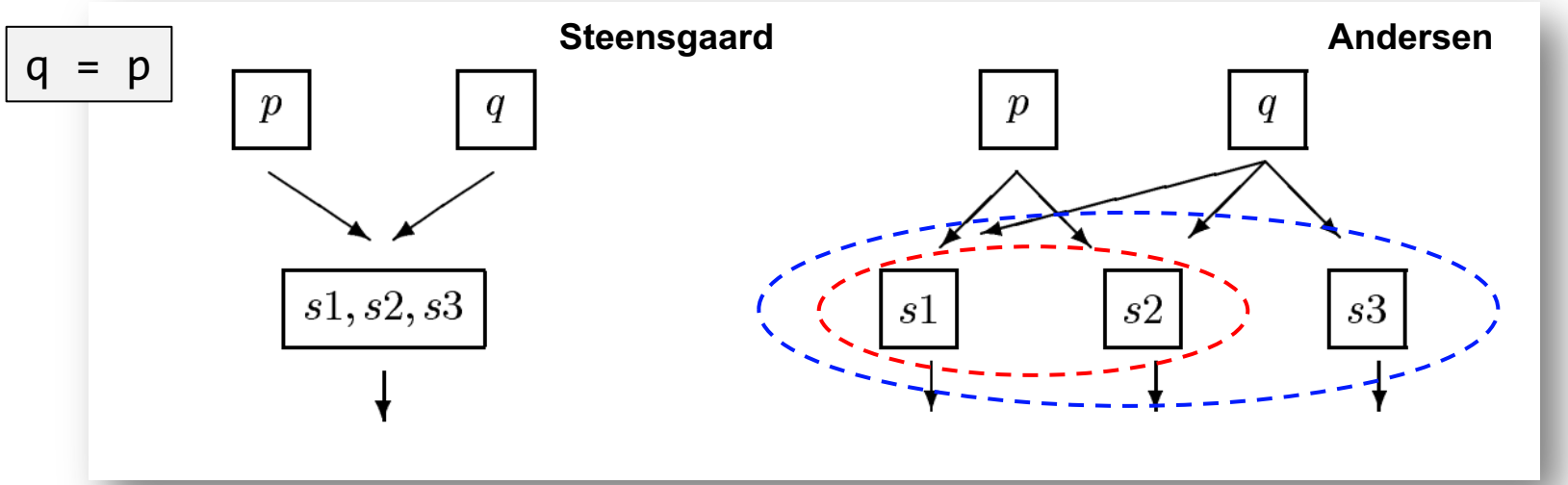
pts(x) = pts(*y)

Disjoint-Set Union-Find





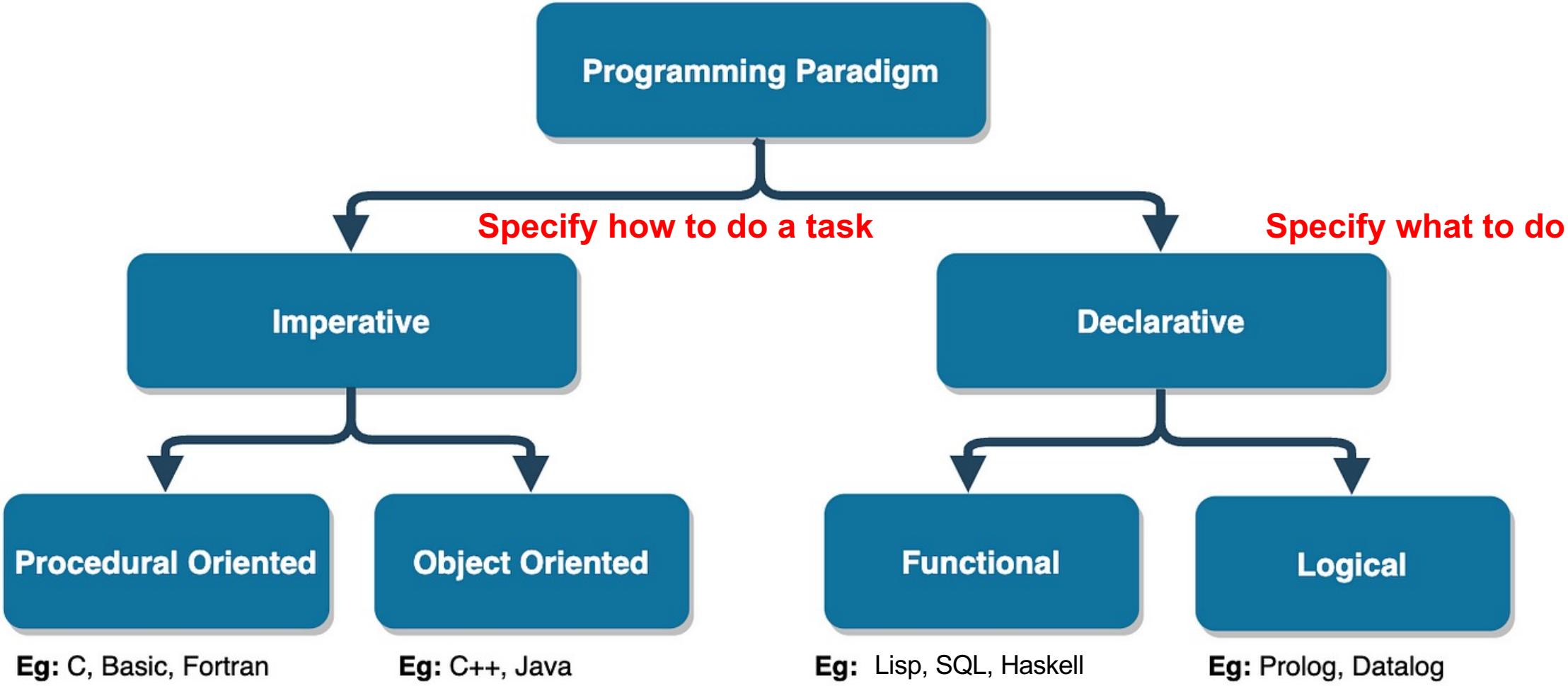
Steensgaard's Algorithm



Steensgaard	Andersen
Sound	Sound
Almost linear (more efficient)	Nearly cubic
Unification-based (less precise)	Inclusion-based

PART IV: Datalog-Based DFA

Recap: Programming Paradigm



Logic Programming

- Logic programming
  - In a broad sense: the use of mathematical logic for programming
- Prolog (1972)
  - Use logical rules to specify how mathematical relations are computed
  - A prolog program is a database of logical rules

Logic Programming

- Example:
  - Nanjing is rainy
  - Beijing is rainy
  - Beijing is cold

  } **Facts**

  - If a city is both rainy and cold, then it is snowy

  } **Rules**

  - **Query:** which city is snowy
  - Search for solutions based on facts and rules

Logic Programming

- Expert systems
- Natural language processing
- Theorem provers
- Reasoning about safety a security
- **Program analysis/Compiler optimization**

What is Datalog?

- Datalog is a subset of Prolog
  - All Datalog programs terminate
  - Ordering of rules does not matter
  - Not Turing complete

- Souffle Datalog engine --- https://souffle-lang.github.io/
  - .decl rainy (c:symbol)
  - .decl cold(c:symbol)
  - .decl snowy(c:symbol)
  - .output snowy
  - rainy(“Nanjing”)
  - rainy(“Beijing”)
  - cold(“Beijing”)
  - snowy(c) :- rainy(c), cold(c)

Predicate/Atom

- Predicates are N-ary relations
  - predicate(x, y, z)
- Examples
  - rainy(“Nanjing”)
  - rainy(“Beijing”)
  - cold(“Beijing”)

Predicate/Atom

- Predicates are N-ary relations
  - predicate(x, y, z)

- Examples
  - rainy(“Nanjing”)
  - rainy(“Beijing”)
  - cold(“Beijing”)
  - brother(x, y) -- x is y’s brother
  - speaks(x, a) -- x speaks language a

Datalog Programming Model

- A Datalog program is a database of Horn clauses
  - h is true **if** the assumptions l_1 l_2 … l_n are simultaneously true
  - **if** --- a sufficient but not necessary condition

$$h :- l_1\ l_2\ \ldots\ l_n$$

- **Example:**
  - rainy(“Nanjing”)
  - snowy(c) :- rainy(c), cold(c)

Datalog Programming Model

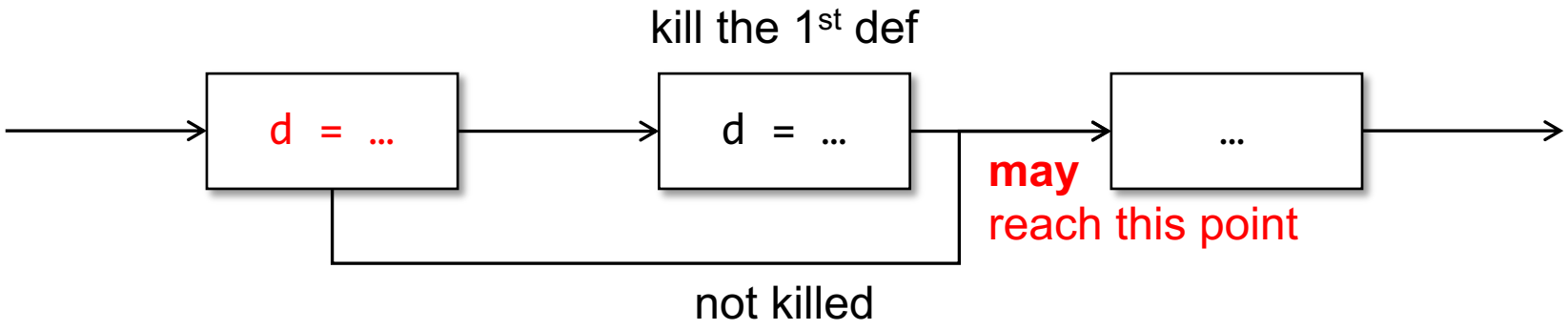
- All rules hold for any instantiation of its variables
  - snowy(c) :- rainy(c), cold(c) for all cities like Nanjing, Beijing, etc.…
- Anything not declared is not true
- Ordering of rules does not matter for results (Prolog vs. Datalog)
- Rules may be recursive
  - reachable(a, b) :- edge(a, b);
  - reachable(a, c) :- edge(a, b), reachable(b, c)

Datalog Programming Model

- Negation is allowed in assumptions
  - more_than_one_hop(a, b) :- reachable(a, b), ¬edge(a, b)
- All rules must be well-formed. Example of bad rules:
  - more_than_one_hop(a, b) :- ¬edge(a, b)
  - **Bad:** a, b on the left do not appear in the positive predicate on the right
  - **Goal:** to ensure termination

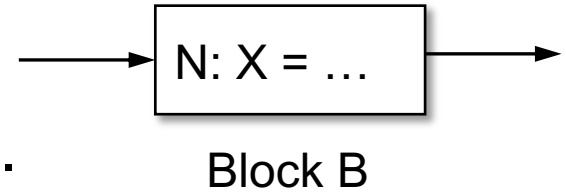
Recap: Reaching Definition

- A definition *d* may reach a program point, if there is a path from *d* to the program point such that *d* is not killed along the path.



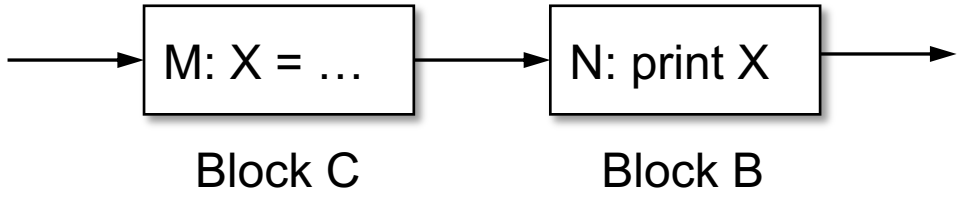
Reaching Definitions by Datalog

- def(B, N, X)
  - the Nth statement in Block B may define the variable X.



- succ(B, N, C)
  - block C is a successor of block B, and B has N statements.

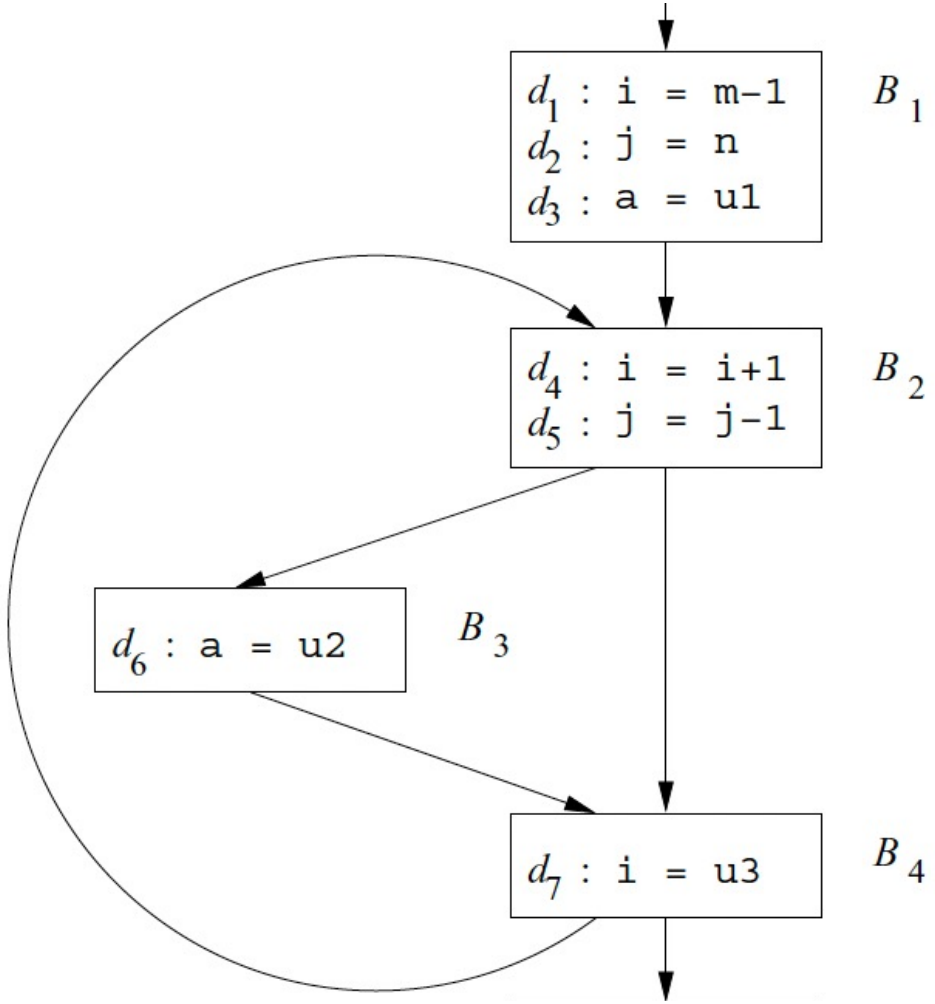
- rd(B, N, C, M, X)
  - the definition of variable X at the Mth statement of block C reaches the Nth statement in B.



Reaching Definitions by Datalog

- rd(B, N, B, N, X) :- def(B, N, X)
- rd(B, N, C, M, X) :- rd(B, N-1, C, M, X), def(B, N, Y), X≠Y
- rd(B, 0, C, M, X) :- rd(D, N, C, M, X), succ(D, N, B)

Reaching Definitions by Datalog



- def(B_1, 1, i)
- def(B_1, 2, j)
- def(B_1, 3, a)
- def(B_2, 1, i)
- def(B_2, 2, j)
- def(B_3, 1, a)
- def(B_4, 1, i)

- succ(B_1, 3, B_2)
- succ(B_2, 2, B_3)
- succ(B_2, 2, B_4)
- succ(B_3, 1, B_4)
- succ(B_4, 1, B_2)

Reaching Definitions by Datalog

- rd(B, N, B, N, X) :- def(B, N, X)
- rd(B, N, C, M, X) :- rd(B, N-1, C, M, X), def(B, N, Y), X≠Y
- rd(B, 0, C, M, X) :- rd(D, N, C, M, X), succ(D, N, B)

- def(B_1, 1, i)
- def(B_1, 2, j)
- def(B_1, 3, a)
- def(B_2, 1, i)
- def(B_2, 2, j)
- def(B_3, 1, a)
- def(B_4, 1, i)
- succ(B_1, 3, B_2)
- succ(B_2, 2, B_3)
- succ(B_2, 2, B_4)
- succ(B_3, 1, B_4)
- succ(B_4, 1, B_2)

Reaching Definitions by Datalog

We just define facts and rules.
Analysis is automatically done by Datalog engines!

Query Example: rd(B_4, 1, B_1, 1, i)

Pointer Analysis by Datalog

- Four kinds of Java statements and ignore procedural call
- **Object creation.** h: T v = **new** T();
- **Copy.** v = w;
- **Field store.** v.f = w;
- **Field load.** v = w.f;

- pts(V, H) variable V can point to a heap object H
- hpts(H, F, G) field F of heap object H can point to heap obj G

Pointer Analysis by Datalog

1) $pts(V, H)$:- "$H : T\ V\ = \texttt{new}\ T$"

2) $pts(V, H)$:- "$V = W$" &
$pts(W, H)$

3) $hpts(H, F, G)$:- "$V.F = W$" &
$pts(W, G)$ &
$pts(V, H)$

4) $pts(V, H)$:- "$V = W.F$" &
$pts(W, G)$ &
$hpts(G, F, H)$

Interprocedural Pointer Analysis

- Dealing with method invocation in Java programs
- x = y.n(z)

- actual(S, I, V): V is the Ith argument in call site S
- formal(M, I, V): V is the Ith parameter in method M

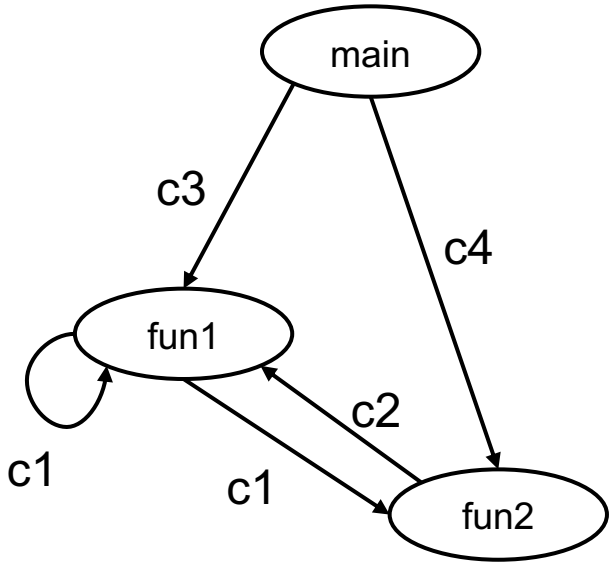
1) $invokes(S, N)$:- "$S: V.N(...)$"

2) $pts(V, H)$:- $invokes(S, M)$ & $formal(M, I, V)$ & $actual(S, I, W)$ & $pts(W, H)$

Just follow the rule for assignment V = W

It is **context-insensitive** as we always do the same thing when calling a function, i.e., do not distinguish different call sites of the same function

Context-Sensitive Pointer Analysis



Call chains:
- c_3-c_1-…
- c_4-c_2-c_1-…
- …

- Assume we already have the call graph.
- Calling context:
  - call chains, paths in the graph
- Context-sensitive analysis:
  - analyze functions in different calling context
- **Difficulty**: Infinite # calling contexts
- **Solution**: restrict the length of call chains

Context-Sensitive Pointer Analysis

- invokes(S, **C**, M, **D**)
- call site S in context C, invokes the method M of context D

call chains of length ≤ a predefined value

- pts(V, **C**, H)
- V points to H in context C

- hpts(H, F, G)
- same as before, field F of H points to G

Context-Sensitive Pointer Analysis

	Context-Insensitive	Context-Sensitive
1	pts(V, H) :- "H : T V = new T()"	pts(V, **C,** H) :- "H : T V = new T()", **invoke(H, C, _, _)**
2	pts(V, H) :- "V = W", pts(W, H)	pts(V, **C,** H) :- "V = W", pts(W, **C,** H)
3	hpts(H, F, G) :- "V.F = W", pts(W, G), pts(V, H)	hpts(H, F, G) :- "V.F = W", pts(W, **C,** G), pts(V, **C,** H)
4	pts(V, H) :- "V = W.F", pts(W, G), hpts(G, F, H)	pts(V, **C,** H) :- "V = W.F", pts(W, **C,** G), hpts(G, F, H)
5	pts(V, H) :- invokes(S, M), formal(M, I, V), actual(S, I, W), pts(W, H)	pts(V, **D,** H) :- invokes(S, **C,** M, **D**), formal(M, I, V), actual(S, I, W), pts(W, **C,** H)

PART V: Object Sensitivity

Object Sensitivity

```
    class List {
        int x;
        void foo() {
c1          print(x);
        }
    }


    class A {
       void bar(List a) {
c2          a.foo();
c3          a.foo();
        }
        …
    }
```

A new form of context-sensitivity, especially for **OO programs**. It does not distinguish contexts based on call sites.

An object-sensitive analysis will not separately analyze $\mathbf{c_2c_1}$ and $\mathbf{c_3c_1}$, because they use the same object a.